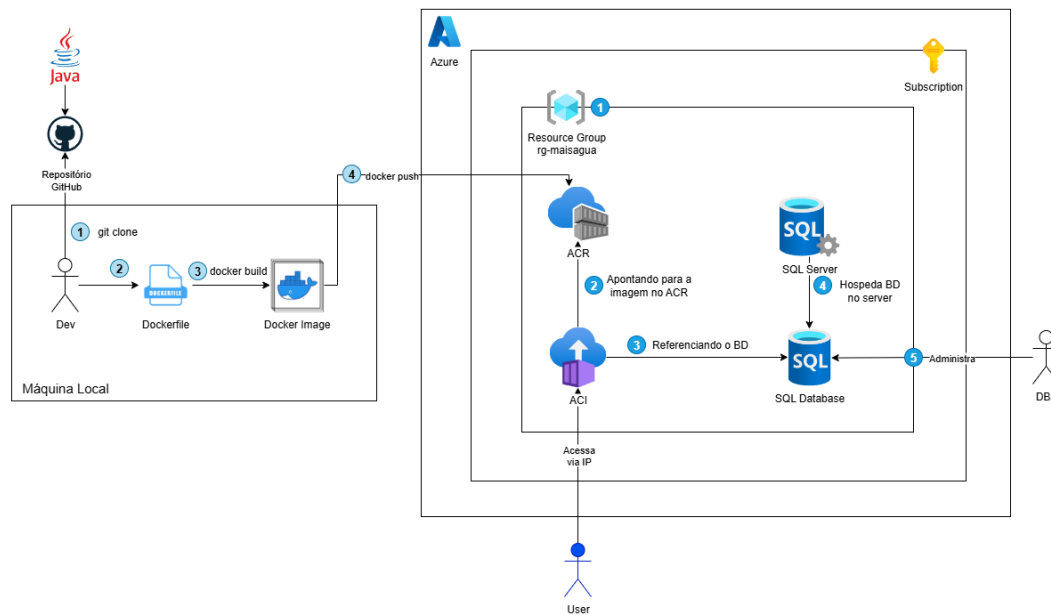


|     |        |       |                               |
|-----|--------|-------|-------------------------------|
| RM: | 558843 | Nome: | Laura de Oliveira Cintra      |
| RM: | 558832 | Nome: | Maria Eduarda Alves da Paixão |
| RM: | 554456 | Nome: | Vinícius Saes de Souza        |

## 1. Arquitetura da Solução

### CP4 - Diagrama

RM 558843 - Laura Cintra  
RM 558832 - Maria Paixão  
RM 554456 - Vinícius Saes



## 2. Configurar um Banco de Dados no Azure SQL Server

### a. Criar grupo de recursos

```
Saes@DESKTOP-1RD6S3G MINGW64 ~/Documents/programacao/Fiap/2_ano/DevOps/2_semestre/3_aula/atividade/MaisAgua-API (main)
$ az group create -l brazilsouth -n rg-maisagua
{
  "id": "/subscriptions/9443a606-9d51-4121-a77c-8bbc27b7a6ea/resourceGroups/rg-maisagua",
  "location": "brazilsouth",
  "managedBy": null,
  "name": "rg-maisagua",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
```

## b. Criar Sql Server

Microsoft Azure

Search resources, services, and docs (617)

Copilot

RM554456@flap.com.br

RM554456@flap.com.br

RM554456@flap.com.br

Create SQL Database Server

Microsoft

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name \*server-maisagua

Location \*South America/Brazil South

Authentication

Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication, or using an existing Microsoft Entra user, group, or application as Microsoft Entra admin, or select both SQL and Microsoft Entra authentication.

Authentication method

☐ Use Microsoft Entra-only authentication

☐ Use both SQL and Microsoft Entra authentication

☒ Use SQL authentication

Server admin login \*admMaisAgua

Password \*

Confirm password \*

## c. Criar banco de dados

Create SQL Database

Microsoft

BasicsNetworkingSecurityAdditional settingsTagsReview + create

Product details

SQL database by Microsoft

Terms of use | Privacy policy

Estimated cost per month

5.93 USD

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription2025-2TDSPI-Vinicius-Saes-RM554456

Resource grouprg-maisagua

RegionBrazil South

Databasename db-maisagua

Server(new) server-maisagua

Authentication methodSQL authentication

Server admin loginadmMaisAgua

Compute + storageBasic: 1 GB storage

Backup storage redundancyLocally-redundant backup storage

SQL

Cost summary

Basic (Basic)

Cost per DTU (in USD)1.19

DTUs selectedx 5

ESTIMATED COST / MONTH5.93 USD

#### d. SQL de criação das tabelas

Query 1 ✕

▶ Run ☐ Cancel query ⬇ Save query ⬇ Export data as ▾ 🍱 Show only Editor

```
1 CREATE TABLE tbl_pais (  
2     id_pais INT IDENTITY(1,1),  
3     nome_pais VARCHAR(50) NOT NULL,  
4     CONSTRAINT pk_pais PRIMARY KEY (id_pais),  
5     CONSTRAINT unq_nome_pais UNIQUE (nome_pais)  
6 );  
7  
8 CREATE TABLE tbl_estado (  
9     id_estado INT IDENTITY(1,1),  
10    nome_estado VARCHAR(50) NOT NULL,  
11    id_pais INT NOT NULL,  
12    CONSTRAINT pk_estado PRIMARY KEY (id_estado),  
13    CONSTRAINT unq_nome_estado UNIQUE (nome_estado),
```

Results Messages

Query succeeded: Affected rows: 0

### 3. Configurar um ACR no Azure

Home > Container registries >

## Create container registry

✓ Validation passed

Basics   Networking   Encryption   Tags   Review + create

### Registry details

#### Basics

|                                            |                                    |
|--------------------------------------------|------------------------------------|
| Registry name                              | crmaisagua                         |
| Subscription                               | 2025-2TDSPJ-Vinicius-Saes-RM554456 |
| Resource Group                             | rg-maisagua                        |
| Location                                   | Brazil South                       |
| Domain name label scope                    | Unsecure                           |
| Availability zones                         | Disabled                           |
| Pricing plan                               | Basic                              |
| Role assignment permissions mode (Preview) | RBAC Registry Permissions          |

#### Networking

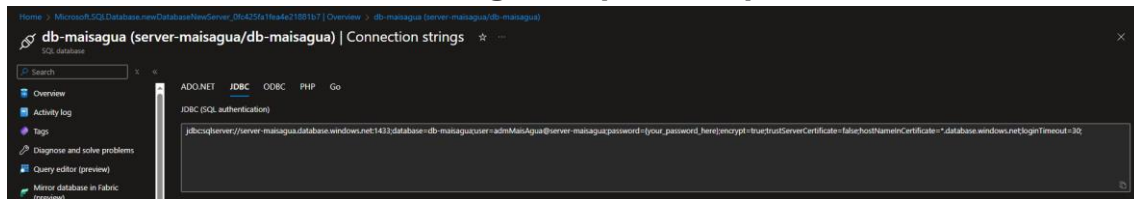
|                       |     |
|-----------------------|-----|
| Public network access | Yes |
|-----------------------|-----|

#### Encryption

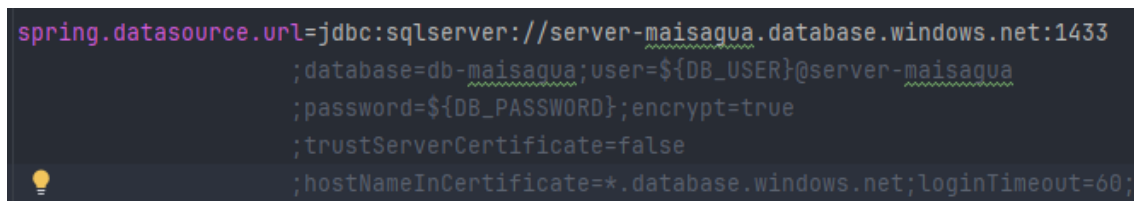
|                      |          |
|----------------------|----------|
| Customer-Managed Key | Disabled |
| Identity             | None     |
| Key Vault            | None     |
| Encryption key       | None     |
| Version              | None     |

## 4. Configurar a Aplicação para Utilizar o Banco de Dados Azure SQL Server

### a. Connection String do Sql server para JDBC

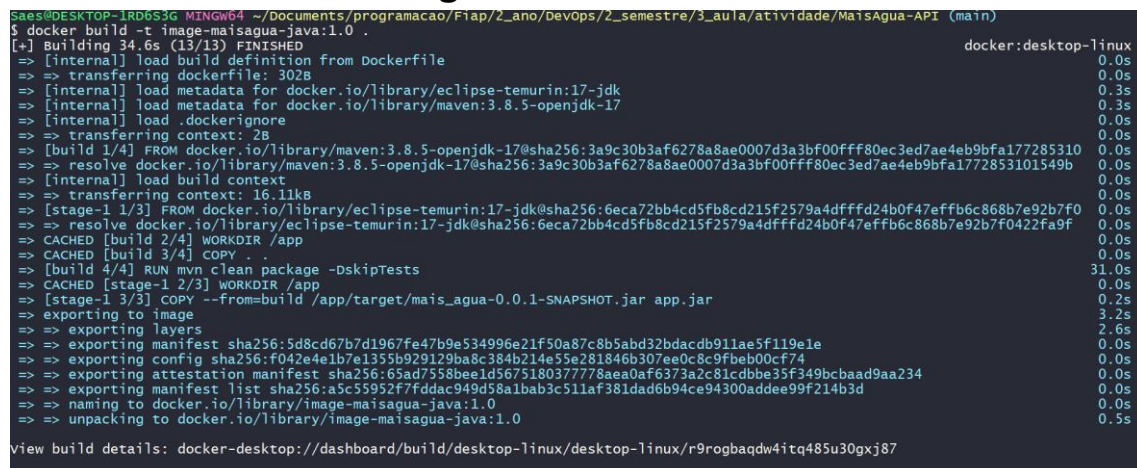


### b. Application Properties do Spring



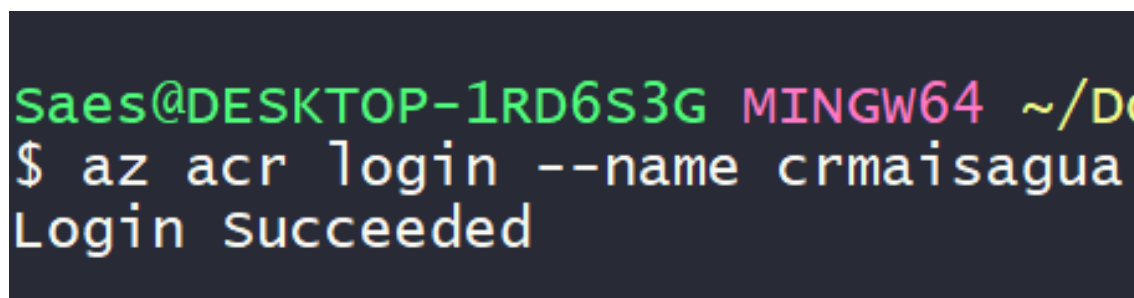
## 5. Criar a Imagem e Realizar o Push da Imagem no ACR

### a. Criando a imagem



### b. Dando push no ACR criado

#### i. Logando no az acr

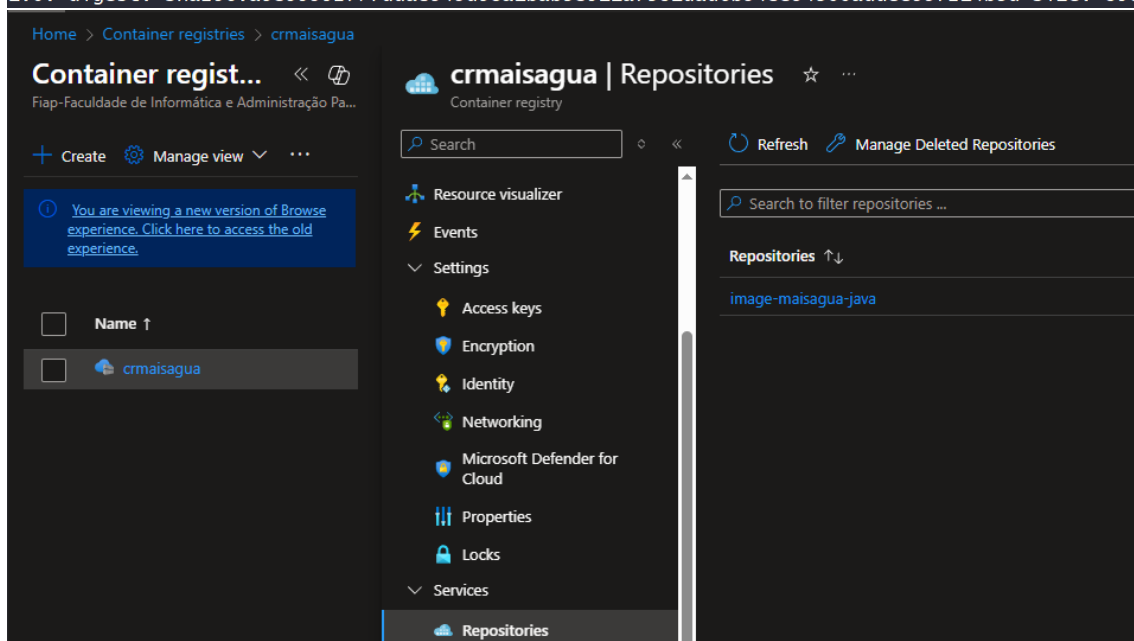


## ii. Preparar tag da imagem

```
Saes@DESKTOP-1RD6S3G MINGW64 ~/Documents/programacao/Fiap/2_ano/DevOps/2_semestre/
$ docker tag image-maisagua-java:1.0 crmaisagua.azurecr.io/image-maisagua-java:1.0
```

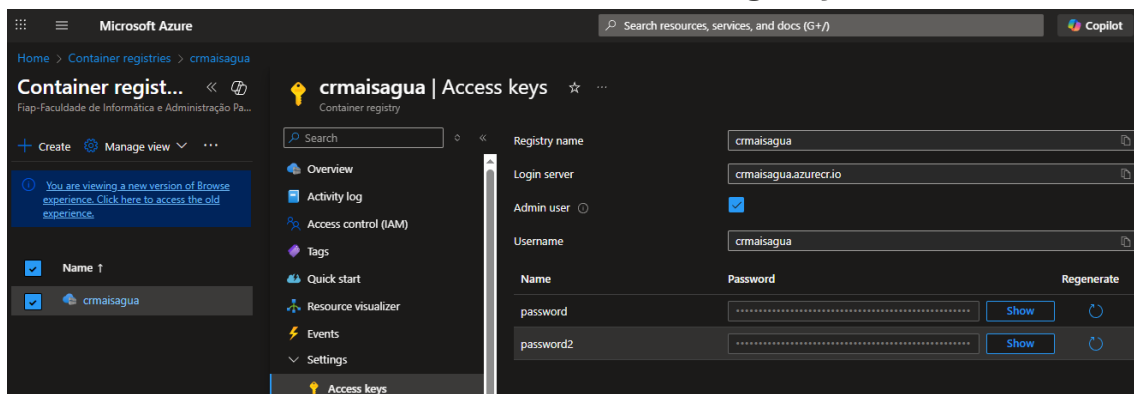
## iii. Publicando no ACR

```
Saes@DESKTOP-1RD6S3G MINGW64 ~/Documents/programacao/Fiap/2_ano/DevOps/2_semestre/3_aula/ativi
$ docker push crmaisagua.azurecr.io/image-maisagua-java:1.0
The push refers to repository [crmaisagua.azurecr.io/image-maisagua-java]
17bc75678de7: Pushed
62e1f75bddef: Pushed
71f0160053e6: Pushed
18b88a7a3d70: Pushed
ced8f9c2b84a: Pushed
cdc74333da4c: Pushed
064c53302191: Pushed
b71466b94f26: Pushed
1.0: digest: sha256:a5c55952f7fddac949d58a1bab3c511af381dad6b94ce94300addee99f214b3d size: 856
```



## 6. Subir a Aplicação no ACI

### a. Habilitando acesso admin ao Registry



## b. Criando o container

# Create container instance

✓ Validation passed

Basics

Networking

Monitoring

Advanced

Tags

Review + create

Basics

Subscription

2025-2TDSPJ-Vinicius-Saes-RM554456

Resource group

rg-maisagua

Region

Brazil South

Container name

container-maisagua-java

SKU

Standard

Image type

Private

Image registry login server

crmaisagua.azurecr.io

Image

crmaisagua.azurecr.io/image-maisagua-java:1.0

Image registry user name

crmaisagua

OS type

Linux

Memory (GiB)

1.5

Number of CPU cores

1

GPU type (preview)

None

GPU count

0

Networking

Networking type

Public

Ports

8080 (TCP)

DNS name label

api-maisagua

DNS name label scope reuse

Tenant

Monitoring

Enable container instance logs

On

Monitoring workspace

DefaultWorkspace-9443a606-9d51-4121-a77c-8bbc27b7a6ea-CQ

Advanced

Restart policy

On failure

Command override

[]

Tags

(none)

 **container-maisagua-java**   

Container instances

 Search

 Start  Restart  Stop  Delete  Refresh  Give feedback

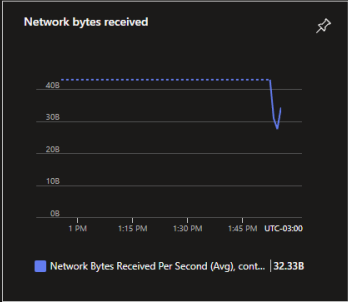
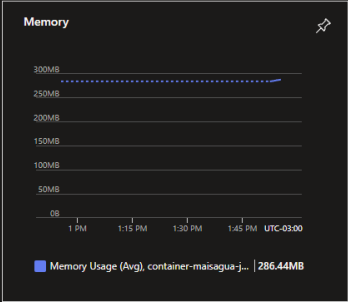
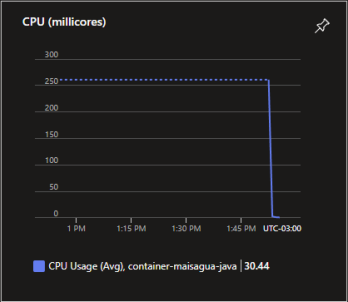
Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Settings
- Monitoring
- Automation
- Help

⚠ Please be aware that Docker Hub has recently introduced a pull rate limit on Docker images. When specifying an image from the Docker Hub registry, this may impact your container instance. [Learn more](#)

Essentials JSON View

|                                       |                                                      |                     |                                                               |
|---------------------------------------|------------------------------------------------------|---------------------|---------------------------------------------------------------|
| Resource group <a href="#">(move)</a> | : <a href="#">rg-maisagua</a>                        | SKU                 | : Standard                                                    |
| Status                                | : Running                                            | OS type             | : Linux                                                       |
| Location                              | : Brazil South                                       | IP address (Public) | : 74.163.174.181                                              |
| Subscription <a href="#">(move)</a>   | : <a href="#">2025-2TDSPJ-Vinicius-Saes-RM554456</a> | FQDN                | : api-maisagua.bwe7ewbceecpehae.brazilsouth.azurecontainer.io |
| Subscription ID                       | : 9443a606-9d51-4121-a77c-8bbc27b7a6ea               | Container count     | : 1                                                           |
| Tags <a href="#">(edit)</a>           | : <a href="#">Add tags</a>                           |                     |                                                               |





## c. Acessando o Swagger da aplicação via ACI

Swagger

v3/api-docs

Explore

+Água API - Sistema de Monitoramento Inteligente

v1 OAS 3.1

Não REST para gerenciamento e monitoramento inteligente de reservatórios de água com foco em eventos extremos como secas e enchentes

Servers

http://74.163.174.181:8080 - Generated server url

Histórico do Reservatório

Operações relacionadas ao monitoramento do nível de água nos reservatórios

GET

/historico-reservatorio/{id}

Buscar histórico por ID

PUT

/historico-reservatorio/{id}

Atualizar histórico

DELETE

/historico-reservatorio/{id}

Excluir histórico

GET

/historico-reservatorio

Listar históricos

POST

/historico-reservatorio

Criar histórico manualmente

Não seguro

4.228.74.11:8080/swagger-ui/index.html#/Usuário/register

Execute

Clear

Responses

Curl

curl -X 'POST' \n'http://4.228.74.11:8080/cadastro' \n-H 'accept: \*/\*' \n-H 'Content-Type: application/json' \n-d '{\n "nome": "saes",\n "email": "saes@teste.com",\n "senha": "12345"\n}'

Request URL

http://4.228.74.11:8080/cadastro

Server response

Code

Details

201

Response body

{\n "idUsuario": 1,\n "nome": "saes",\n "email": "saes@teste.com"\n}

Response headers

cache-control: no-cache,no-store,max-age=0,must-revalidate\nconnection: keep-alive\ncontent-type: application/json\ndate: Tue,26 Aug 2025 17:10:52 GMT\nexpires: 0\nkeep-alive: timeout=60\npragma: no-cache\ntransfer-encoding: chunked\nvary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers\nx-content-type-options: nosniff\nx-frame-options: DENY\nx-xss-protection: 0

Home > db-maisagua (server-maisagua/db-maisagua)

db-maisagua (server-maisagua/db-maisagua) | Query editor (preview) ☆ ...

SQL database

Search

Login + New Query ↑ Open query Feedback Getting started

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

Settings

Data management

Integrations

Power Platform

Security

Intelligent performance

Monitoring

Automation

Help

Resource health

Support + Troubleshooting

db-maisagua (RM554456@fiap.com.br)

Showing limited object explorer here. For full capability please click here to open Azure Data Studio.

Tables

- dbo.tbl\_alerta
- dbo.tbl\_cidade
- dbo.tbl\_clima\_resumo\_sem
- dbo.tbl\_dispositivo
- dbo.tbl\_endereco
- dbo.tbl\_estado
- dbo.tbl\_historico\_reservatorio
- dbo.tbl\_leitura\_dispositivo
- dbo.tbl\_pais
- dbo.tbl\_reservatorio
- dbo.tbl\_reservatorio\_dispositivo
- dbo.tbl\_status\_alerta
- dbo.tbl\_status\_reservatorio
- dbo.tbl\_unidade
- dbo.tbl\_usuario

Query 1

Run Cancel query Save query Export data as Show only Editor

```
1 SELECT * FROM tbl_usuario;
```

Results Messages

Search to filter items...

| id_usuario | nome | email          | senha                                           |
|------------|------|----------------|-------------------------------------------------|
| 1          | saes | saes@teste.com | \$2a\$10\$1vslgahHXQLQ1Dpo3cOZvaLYuGNacM/mH1... |

## 7. Limpando os recursos

```
Saes@DESKTOP-1RD6S3G MINGW64 ~/Documents/programacao/Fiap/
Agua-API (main)
$ az group delete --resource-group rg-maisagua
Are you sure you want to perform this operation? (y/n): y
```